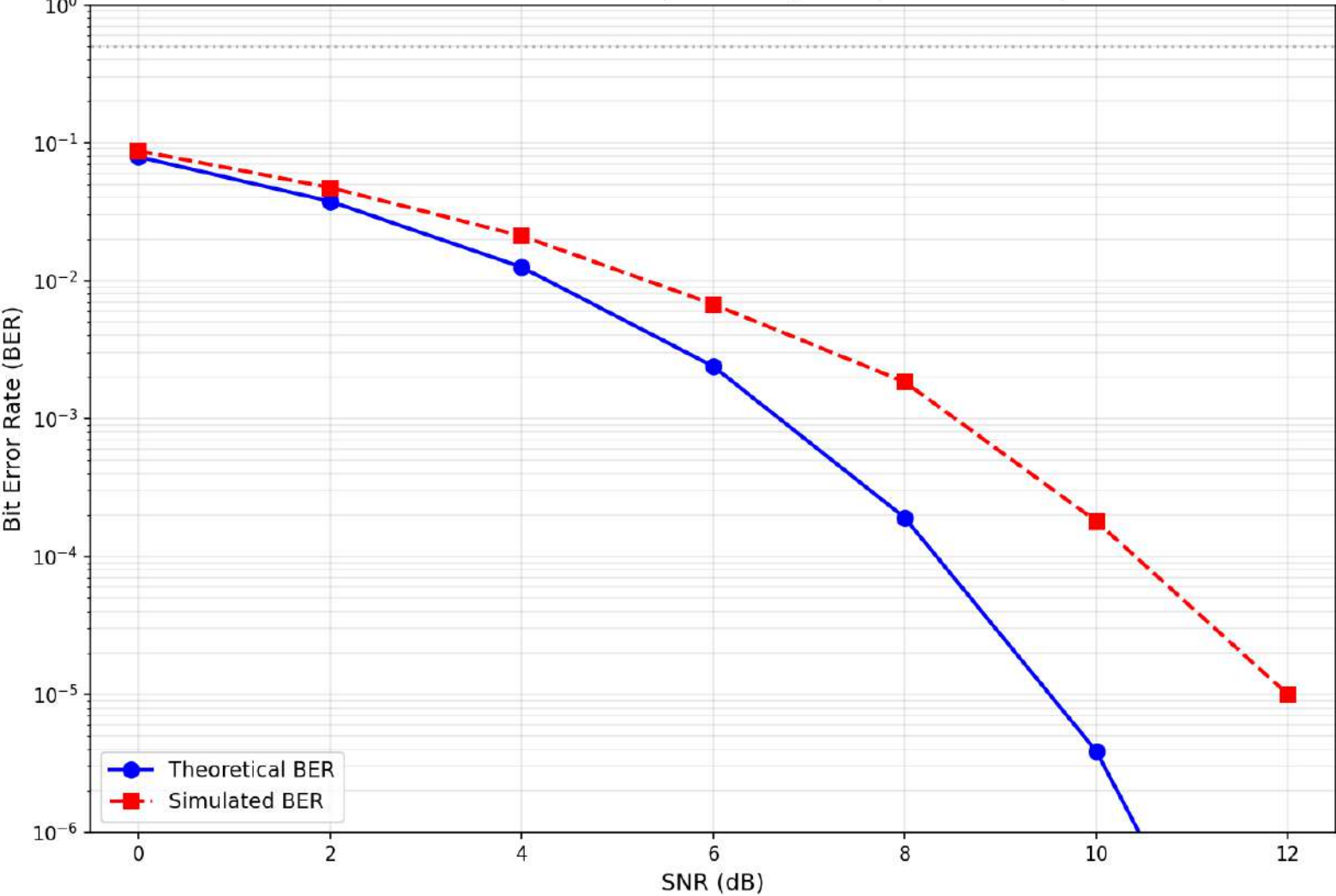
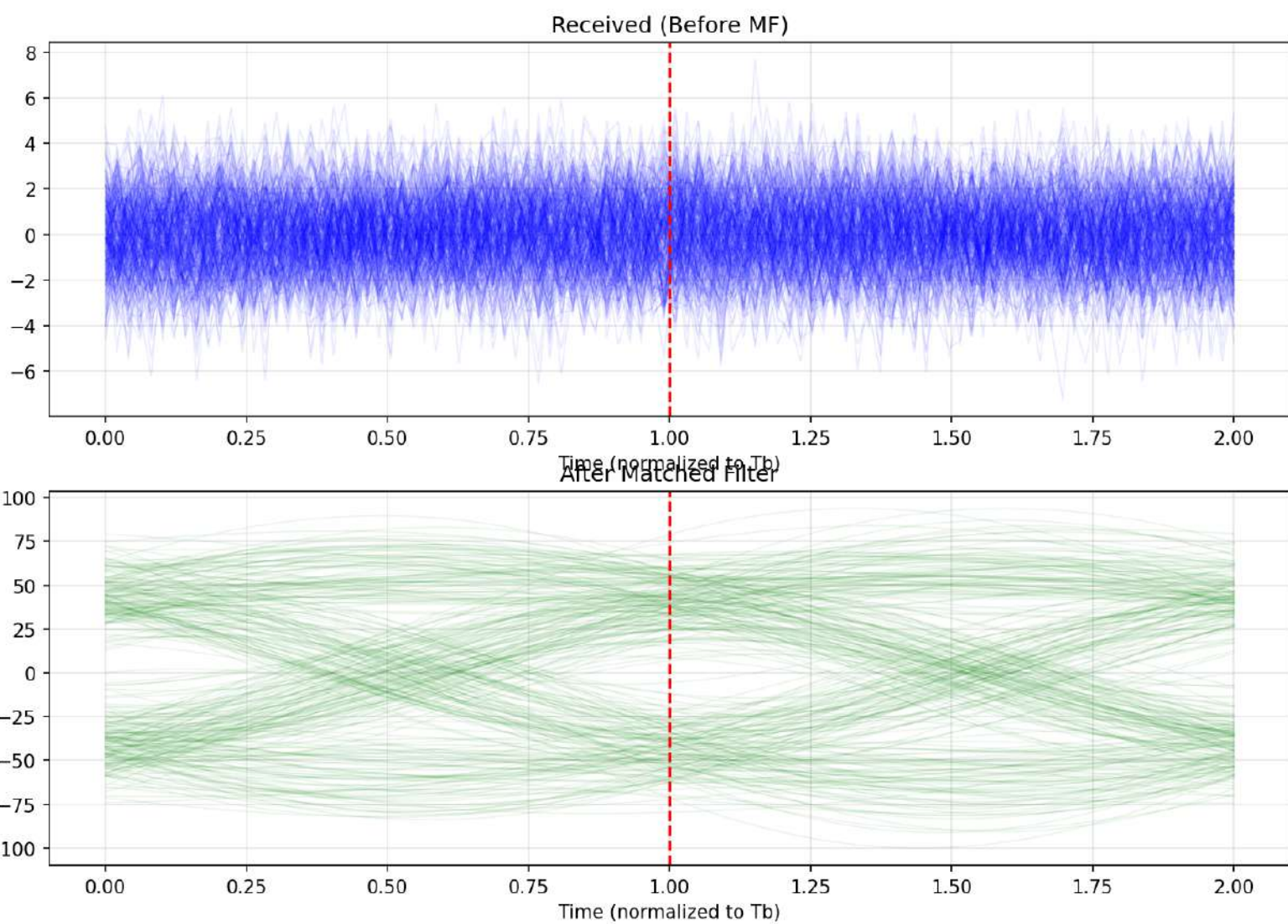


BER vs SNR Waterfall Curve | RC Pulse $\beta=0.5$ | 100,000 bits/point

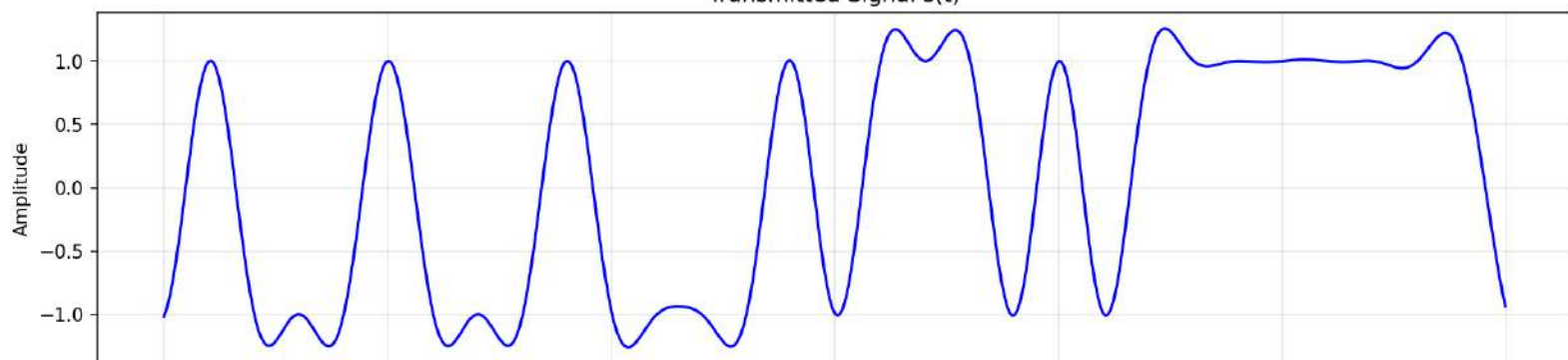


Eye Diagram: Before vs After Matched Filter | SNR=10.0 dB



Block 5 | Signal at Different Stages | SNR = 10.0 dB

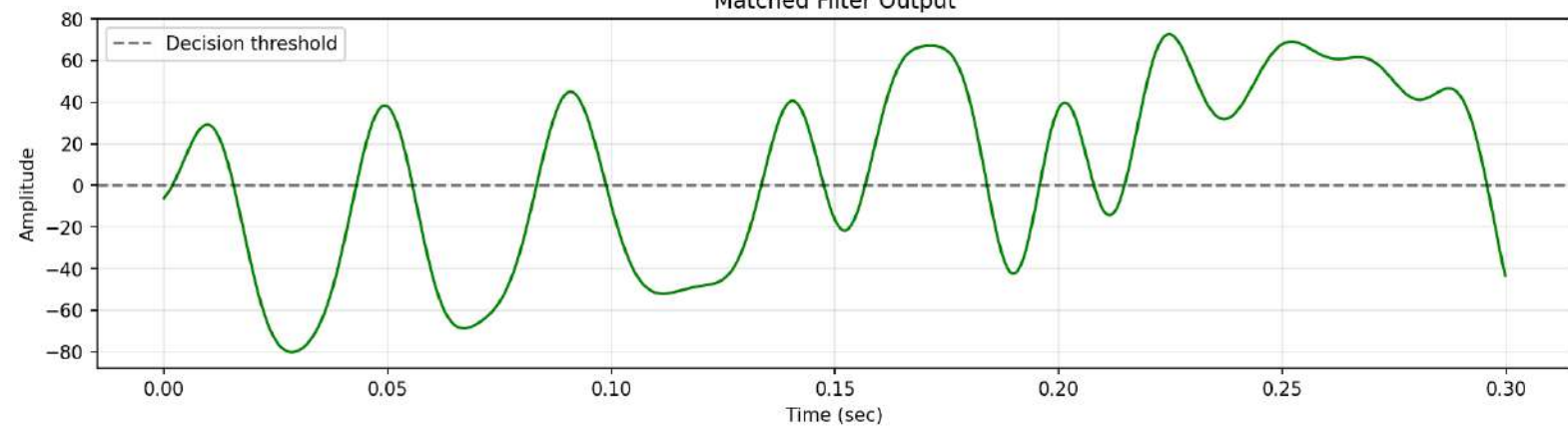
Transmitted Signal $s(t)$



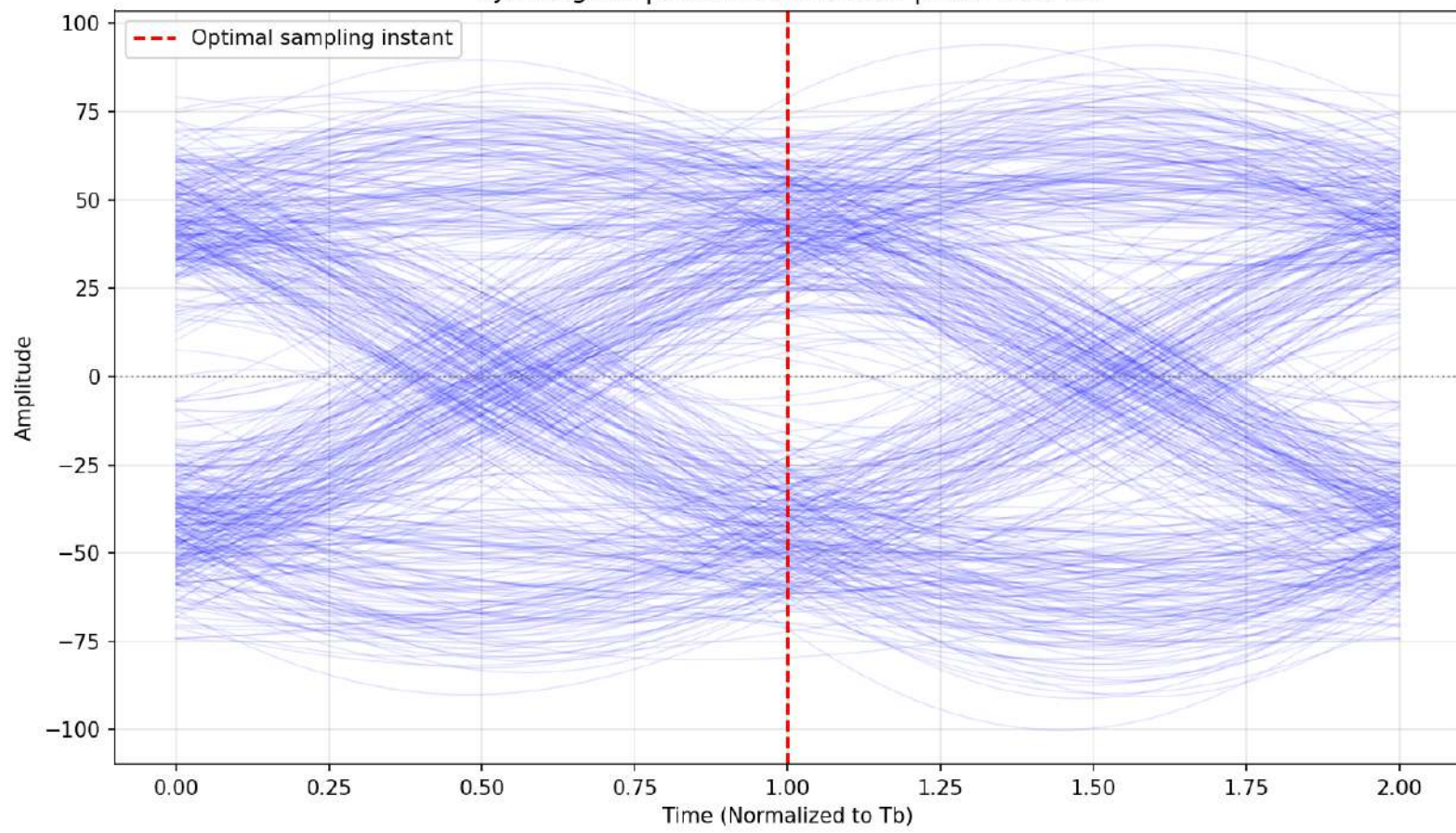
Received Signal $r(t) = s(t) + \text{noise}$ [SNR = 10.0 dB]



Matched Filter Output

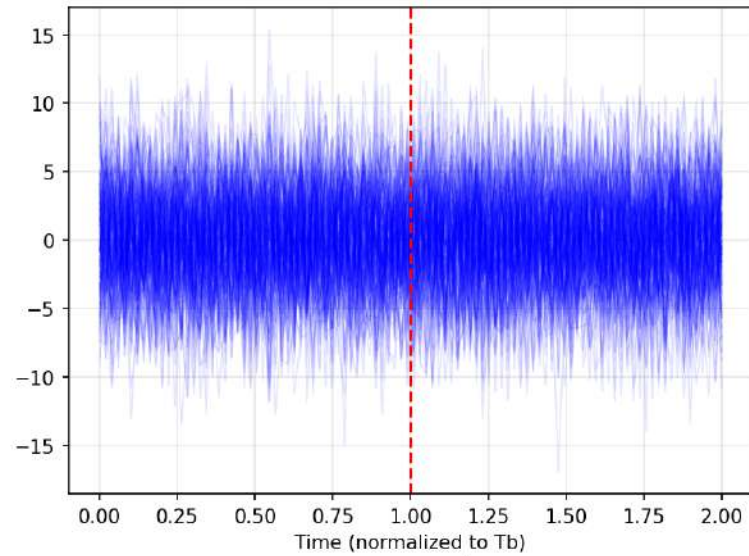


Eye Diagram | After Matched Filter | SNR=10.0 dB

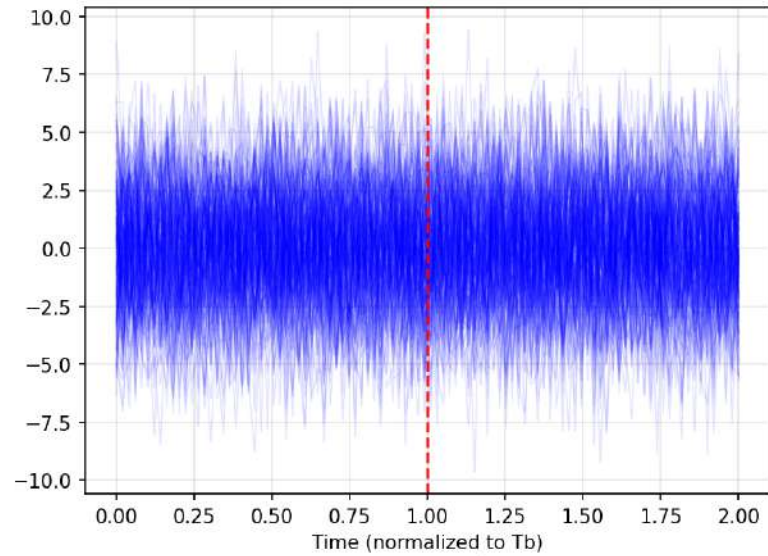


Eye Diagrams at Different E_b/N_0 Levels (Before Matched Filter)

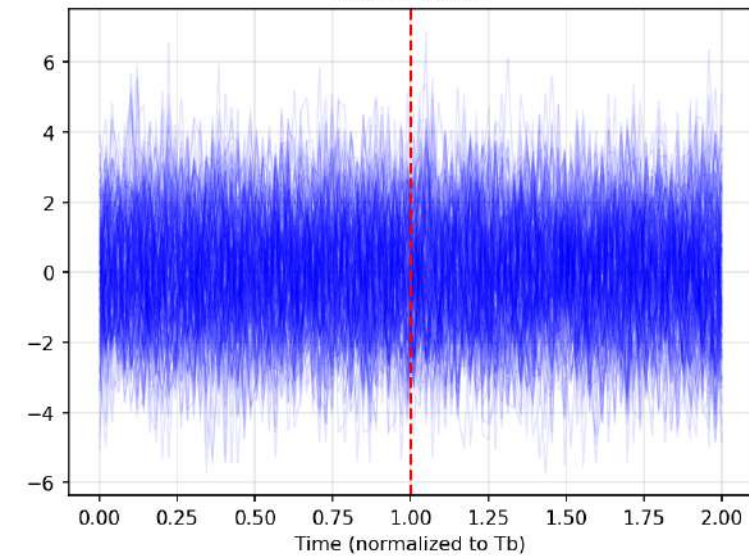
SNR = 2 dB



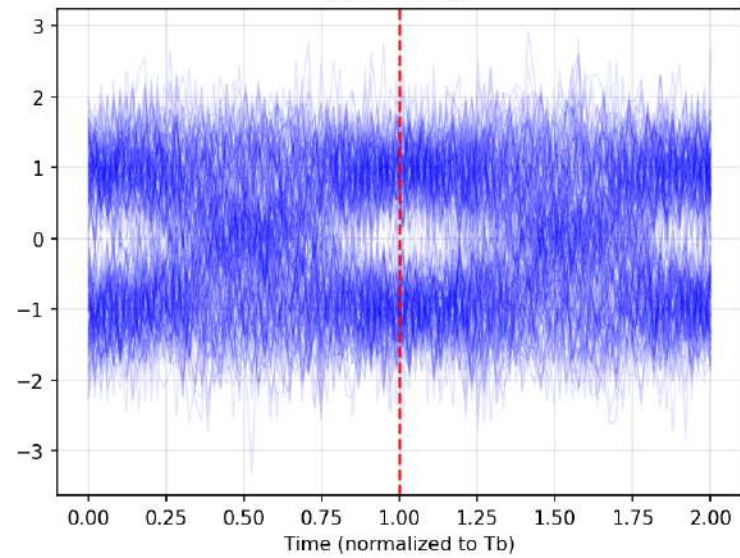
SNR = 6 dB



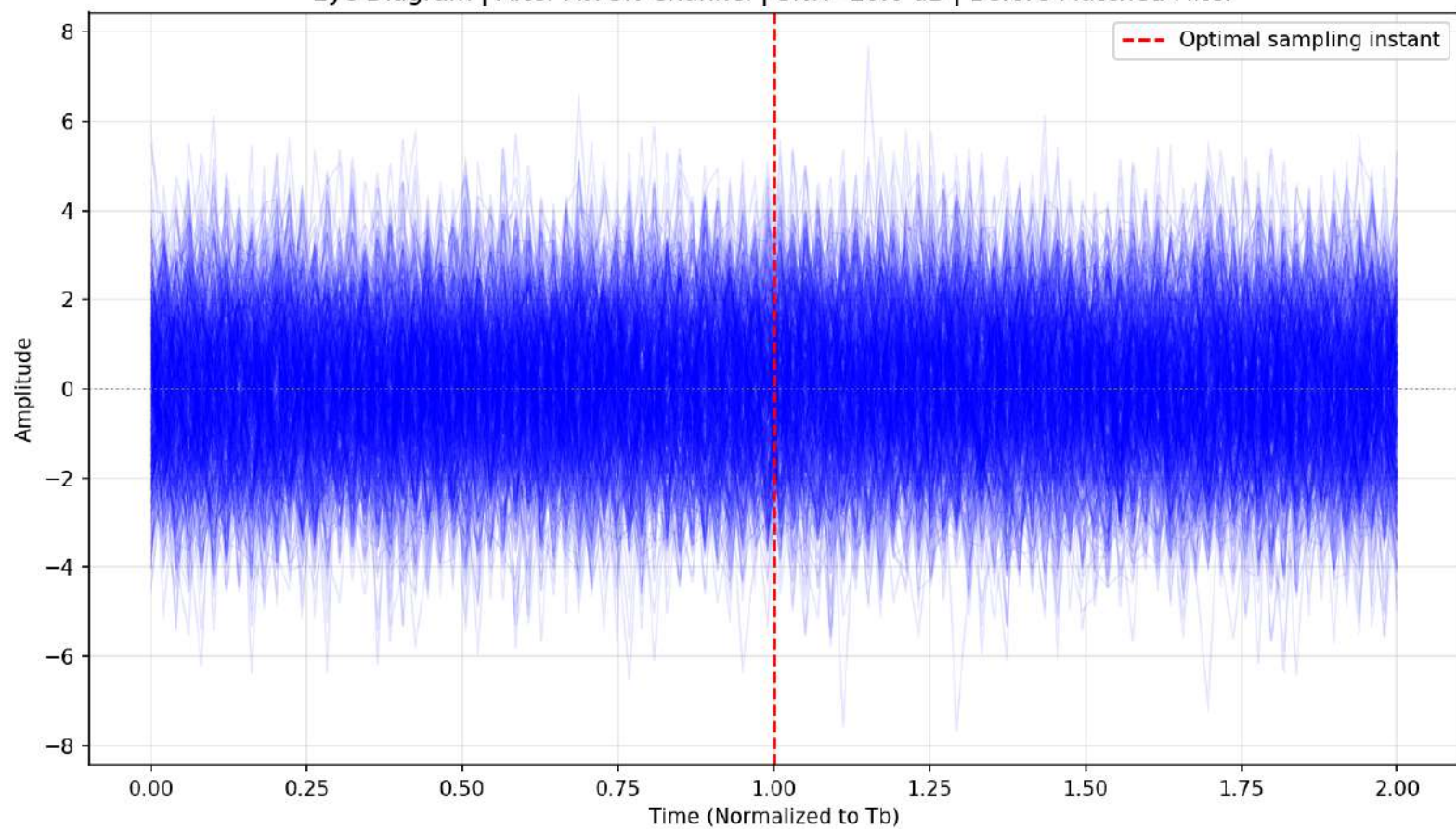
SNR = 10 dB



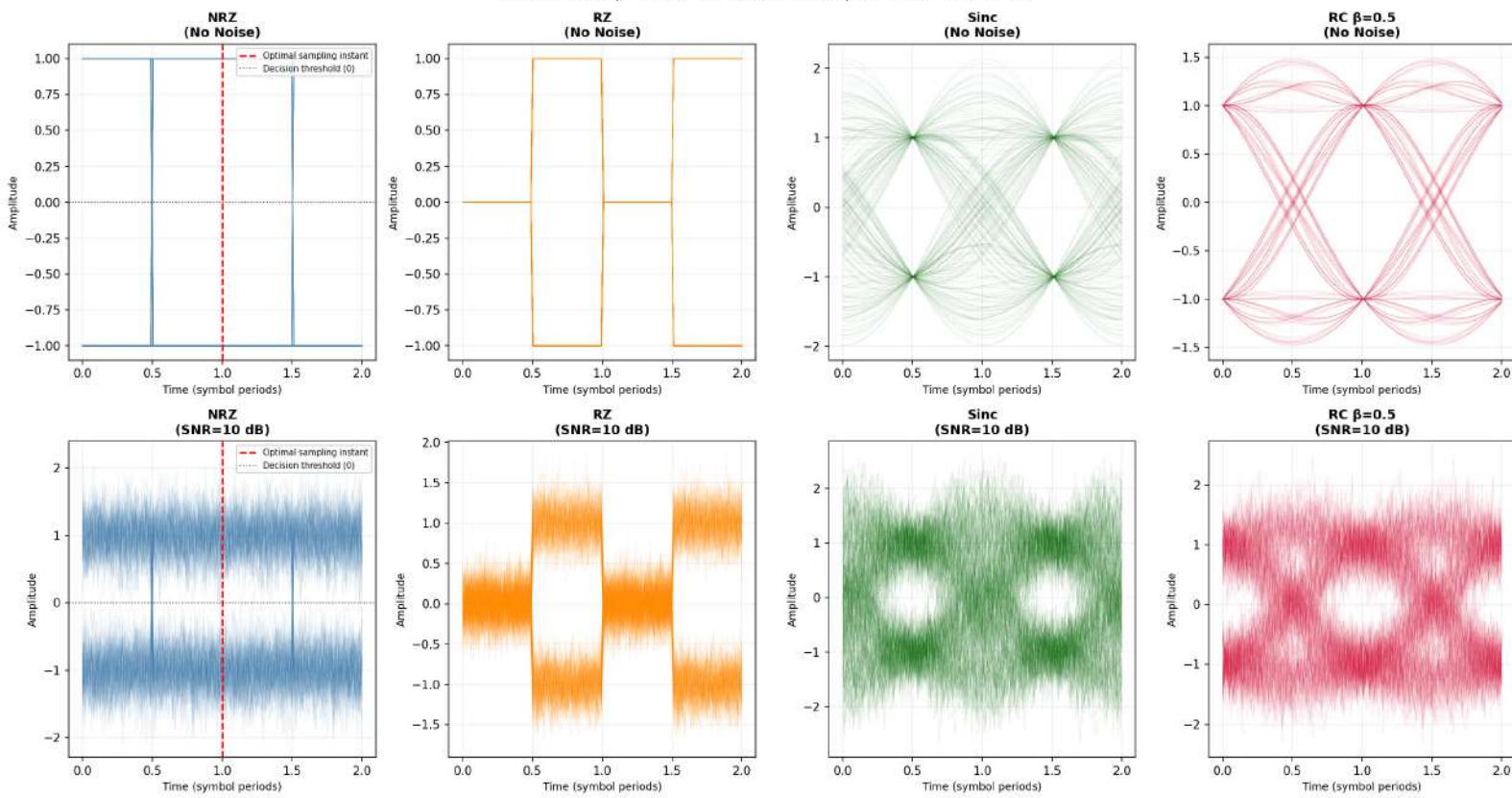
SNR = 20 dB



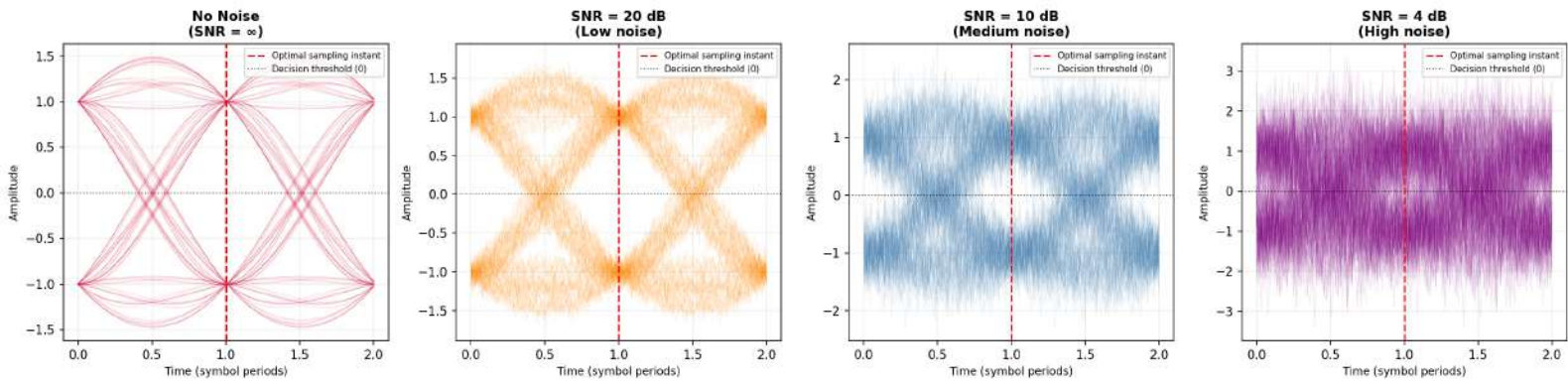
Eye Diagram | After AWGN Channel | SNR=10.0 dB | Before Matched Filter



Eye Diagrams — All Pulse Types
Top: No Noise | Bottom: AWGN SNR=10 dB | Tau=64, 400 symbols

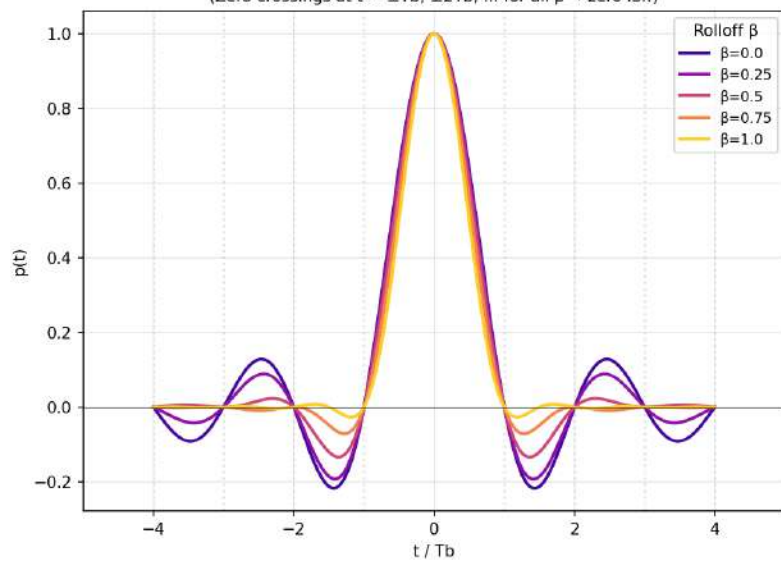


Eye Diagram: Effect of AWGN Noise on RC Pulse ($\beta=0.5$)
Noise closes the eye $\rightarrow$ reduces noise margin

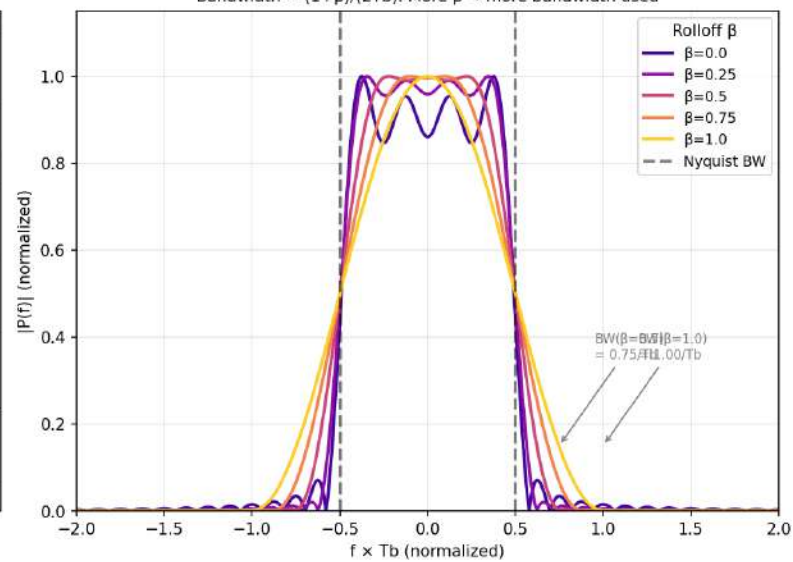


Raised Cosine Pulse: Effect of Rolloff Factor β $\beta=0 \rightarrow$ Sinc (minimum bandwidth, hardest) | $\beta=1 \rightarrow$ maximum rolloff (easiest)

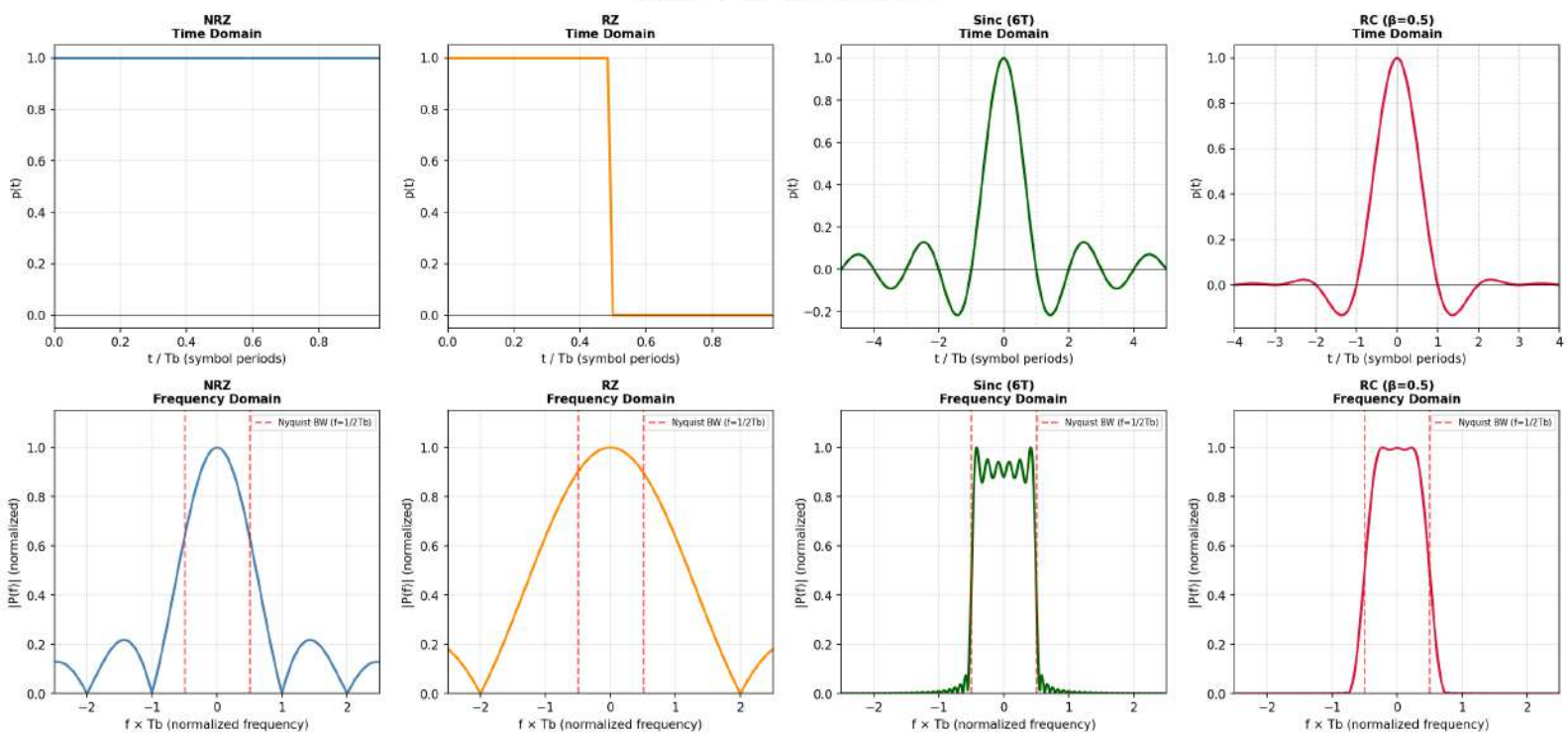
RC Pulse: Time Domain
 (Zero crossings at $t = \pm T_b, \pm 2T_b, \dots$ for all $\beta \rightarrow$ zero ISI!)



RC Pulse: Frequency Domain
 Bandwidth = $(1+\beta)/(2T_b)$. More $\beta \rightarrow$ more bandwidth used

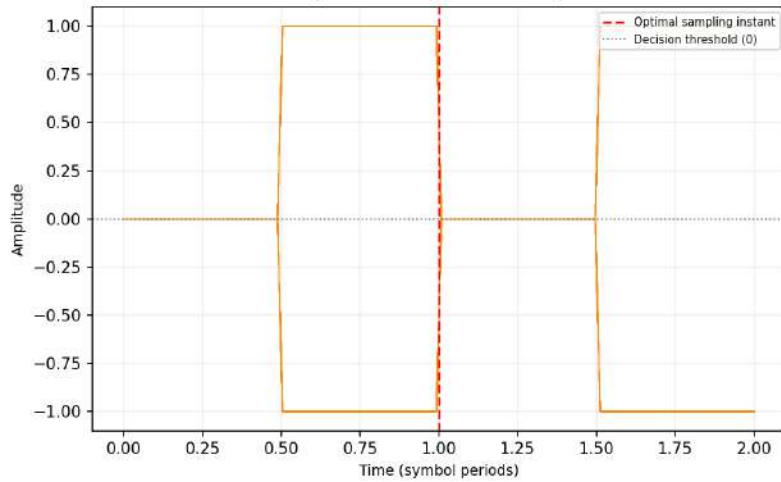


Pulse Shape Comparison: Time Domain (top) and Frequency Domain (bottom)
All amplitudes normalized to 1 at $t=0$

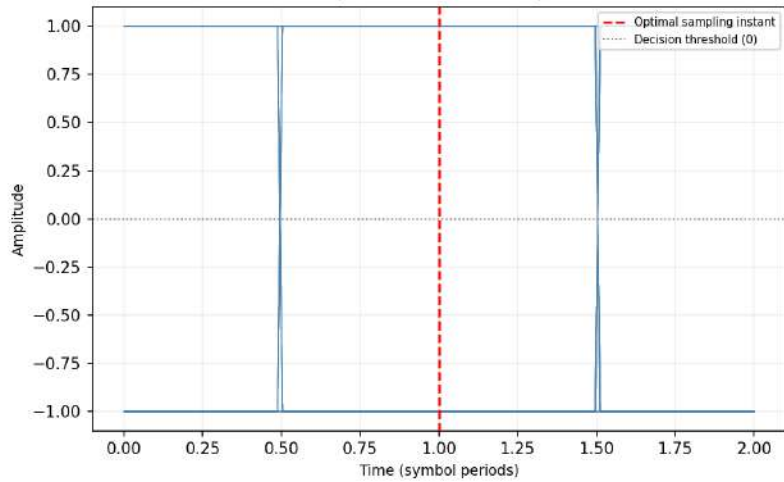


Binary Eye Diagrams (Section 7.10)
(400 random bits, $\tau=64$ samples/symbol, RC $\beta=0.5$)

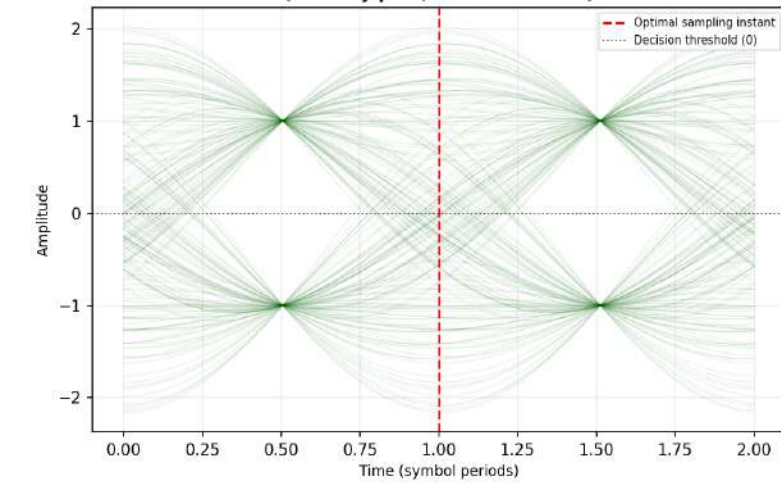
RZ Eye Diagram
(Returns to zero each bit)



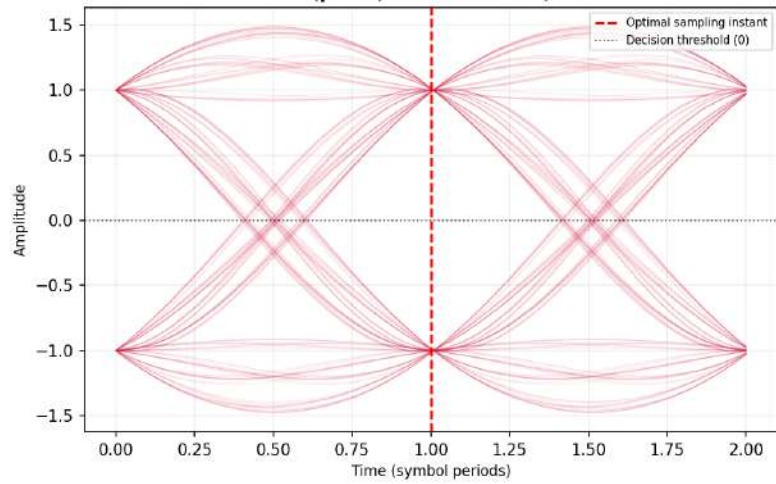
NRZ Eye Diagram
(Non-Return-to-Zero)

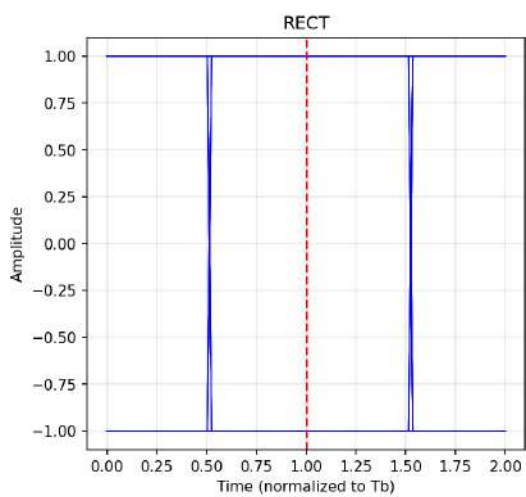


Sinc Pulse Eye Diagram
(Ideal Nyquist, truncated to $6T$)

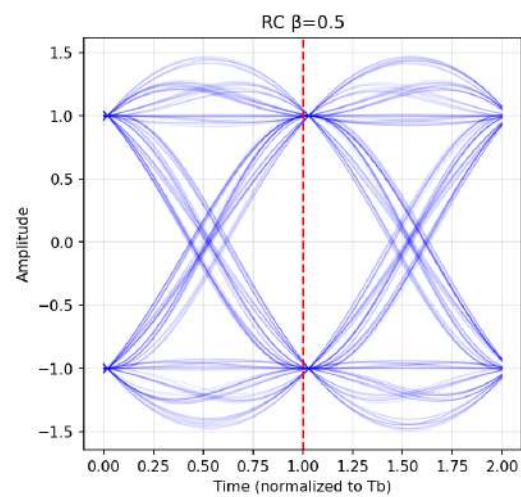
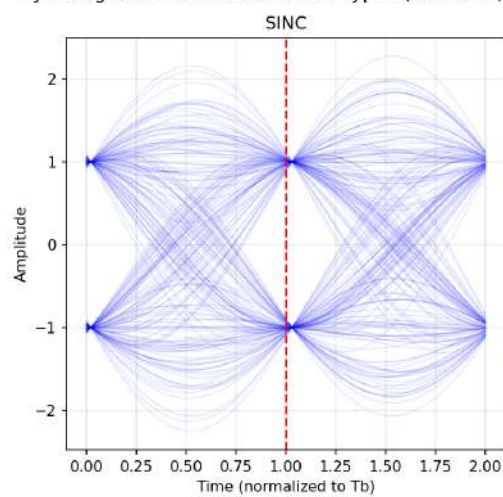


Raised Cosine Eye Diagram
($\beta=0.5$, truncated to $4T$)

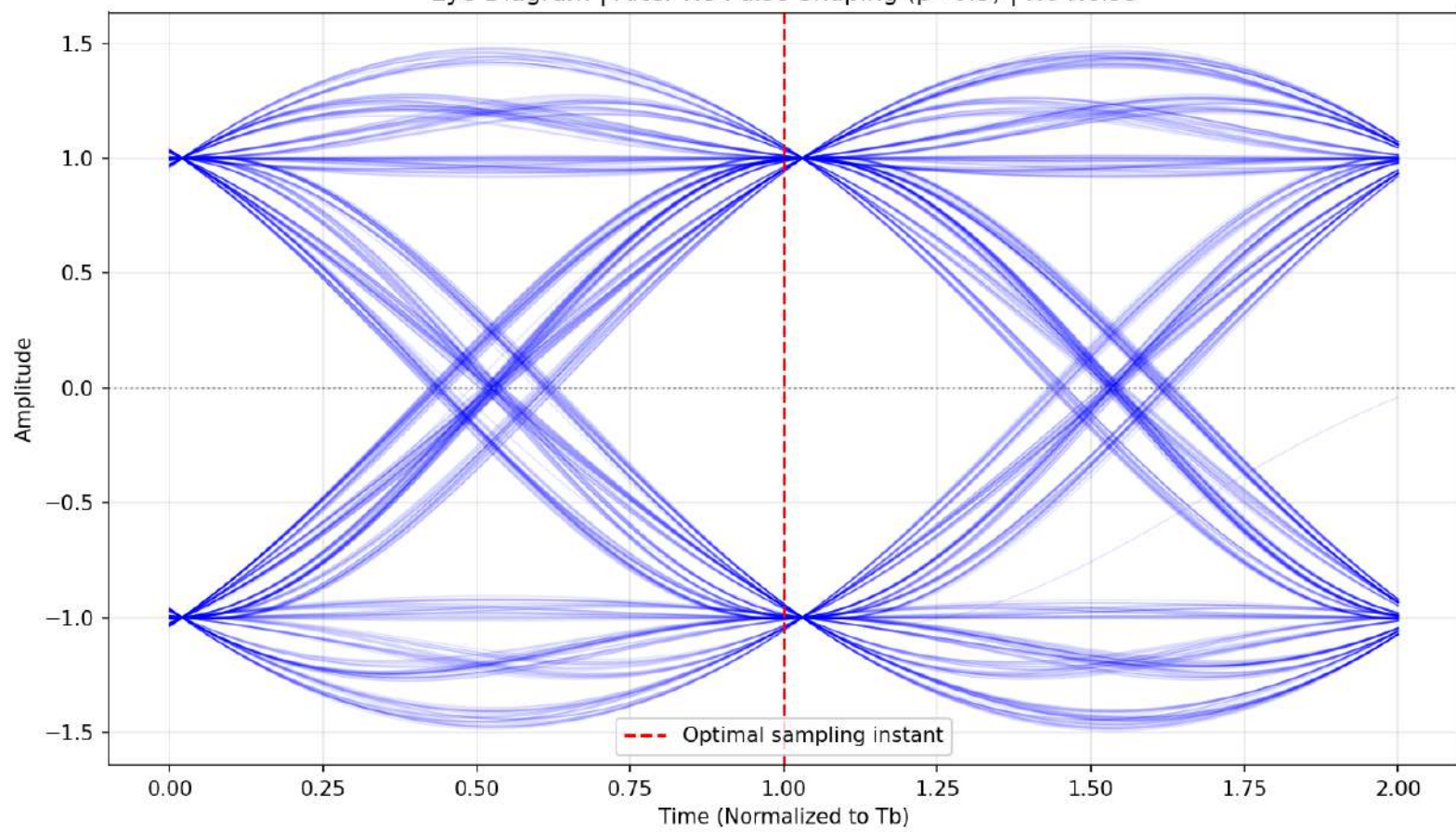




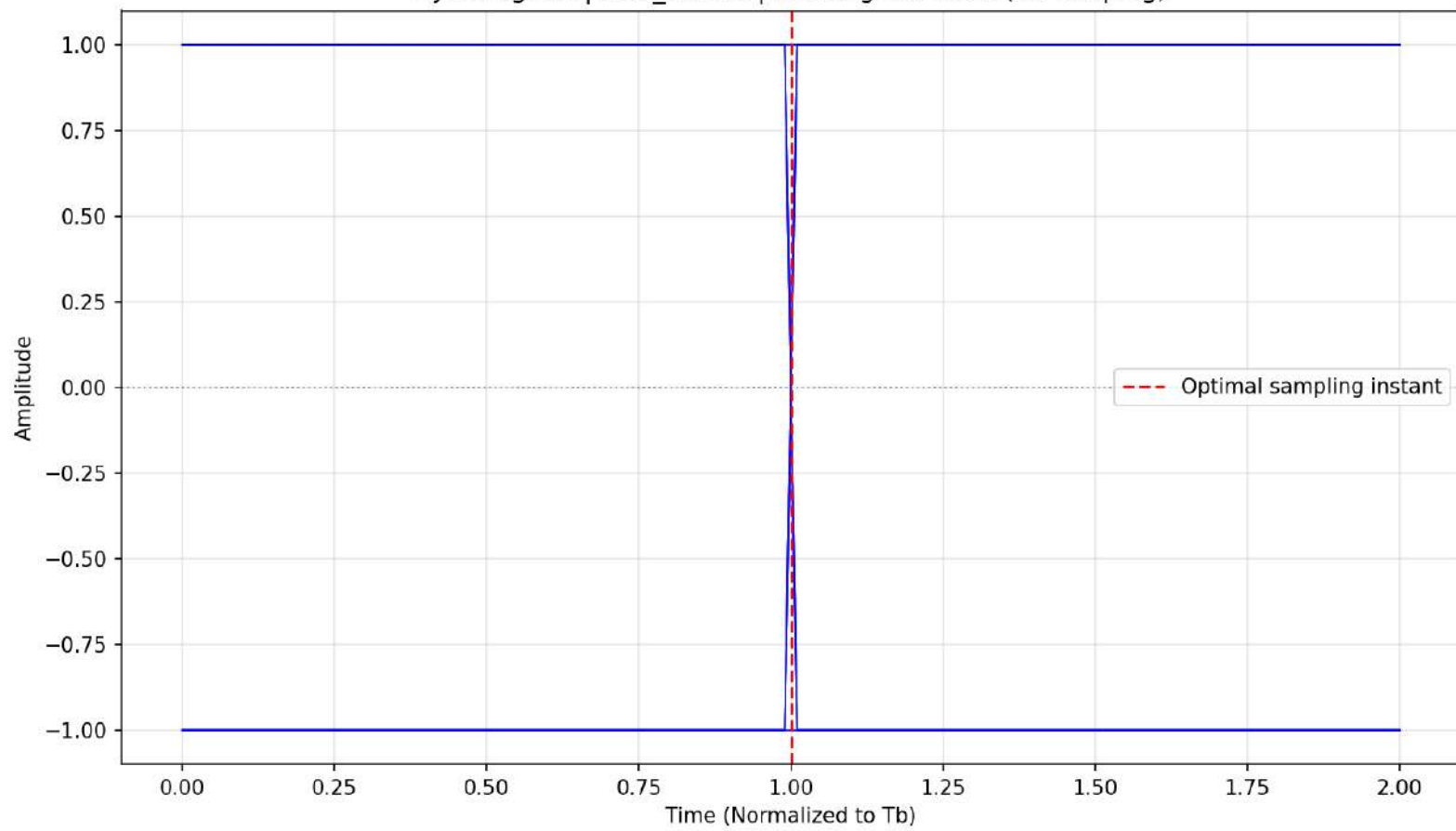
Eye Diagrams for Different Pulse Types (No Noise)



Eye Diagram | After RC Pulse Shaping ($\beta=0.5$) | No Noise

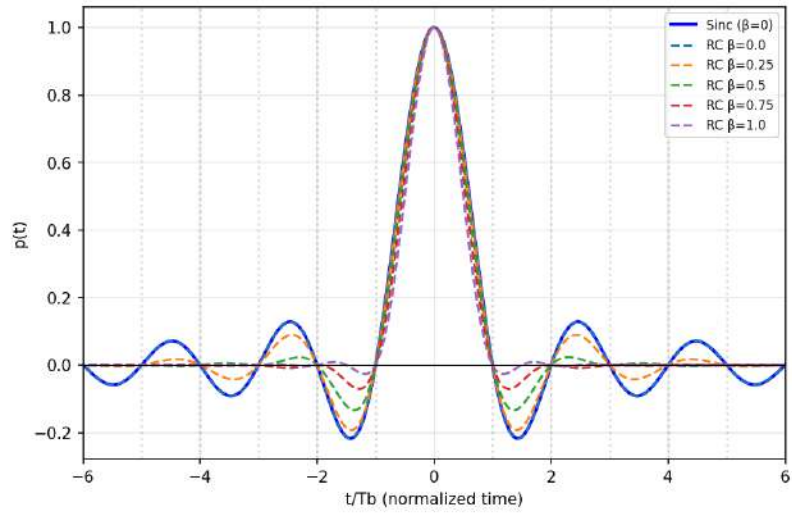


Eye Diagram | NRZ_POLAR | Rectangular Pulse (No Shaping)

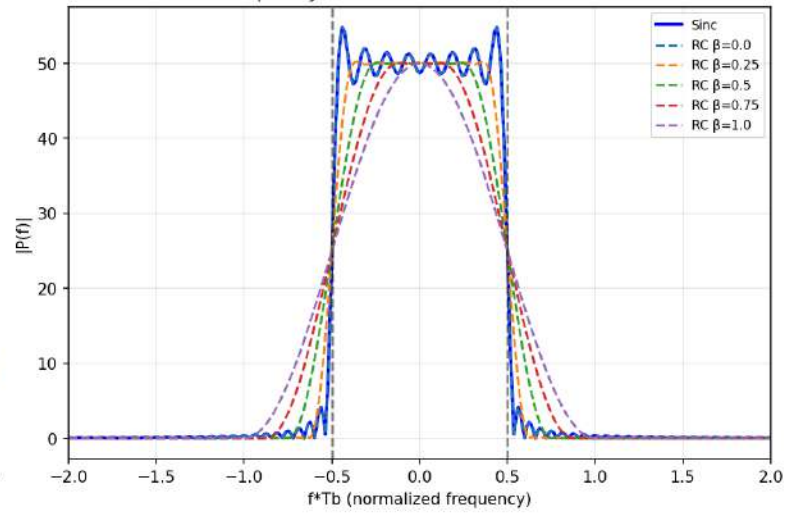


Pulse Shape Comparison: Rectangular, Sinc, and Raised Cosine

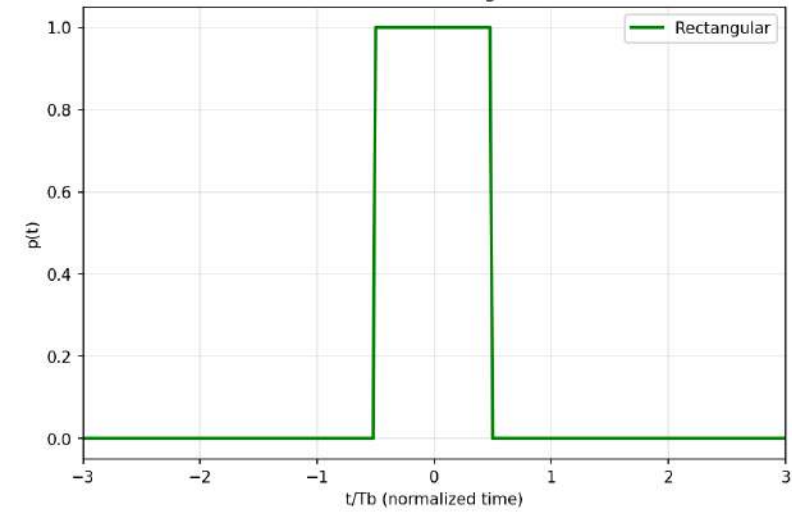
Time Domain: Sinc vs Raised Cosine



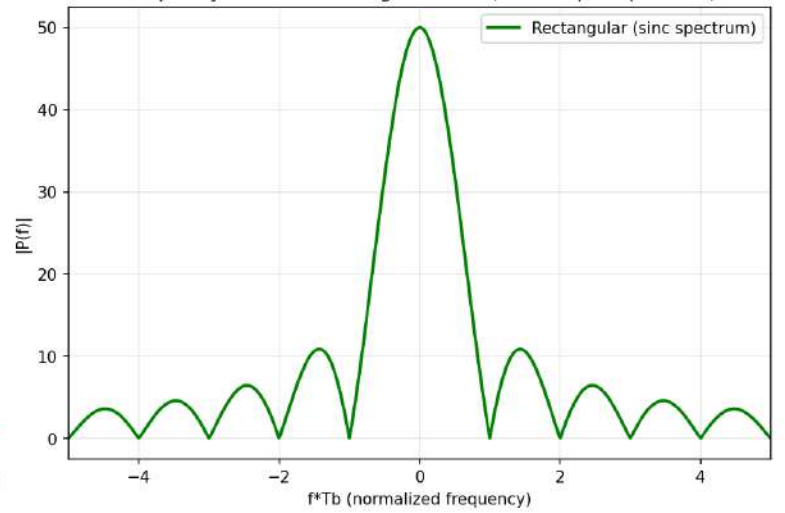
Frequency Domain: Sinc vs Raised Cosine



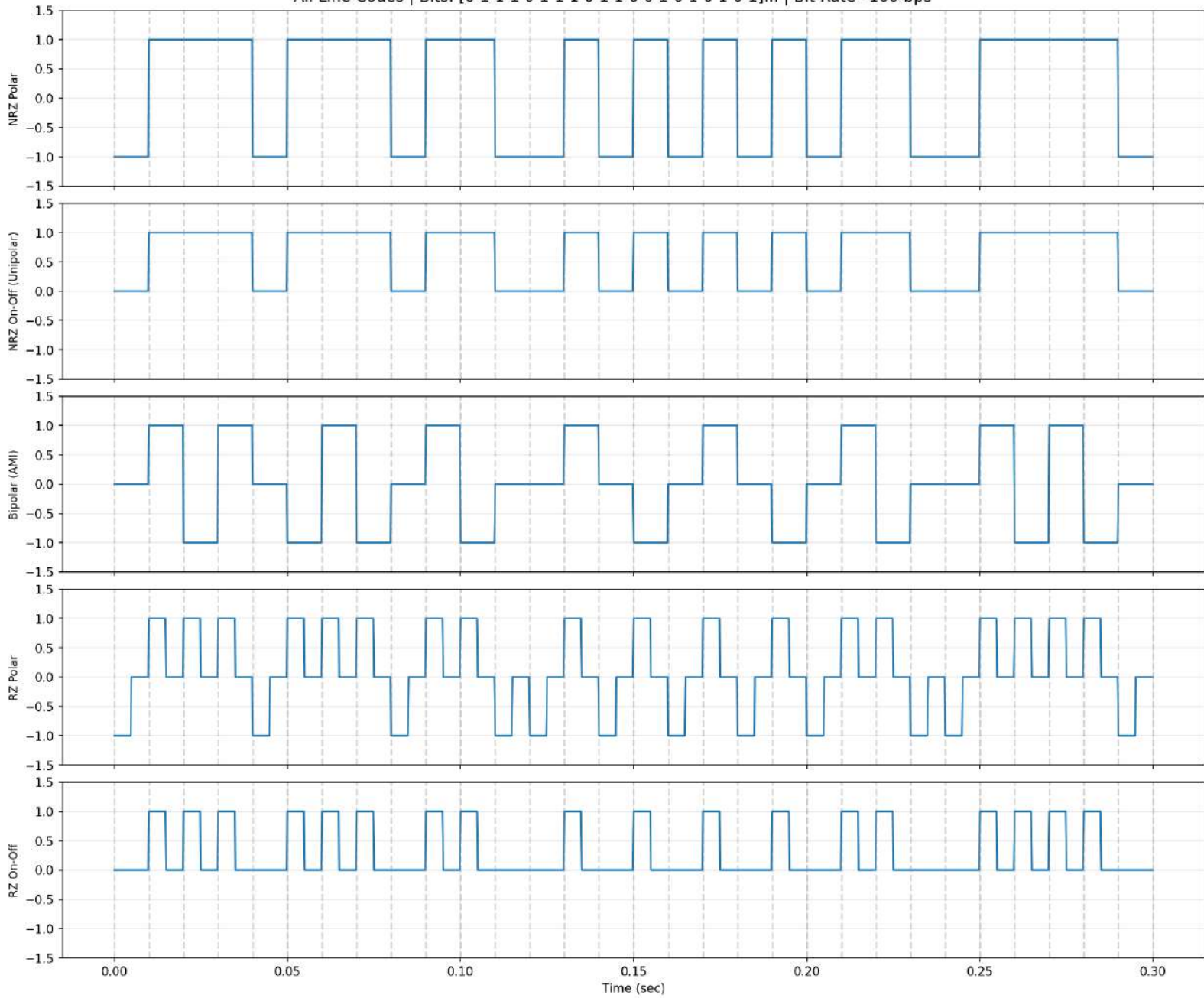
Time Domain: Rectangular Pulse



Frequency Domain: Rectangular Pulse (sinc-shaped spectrum)

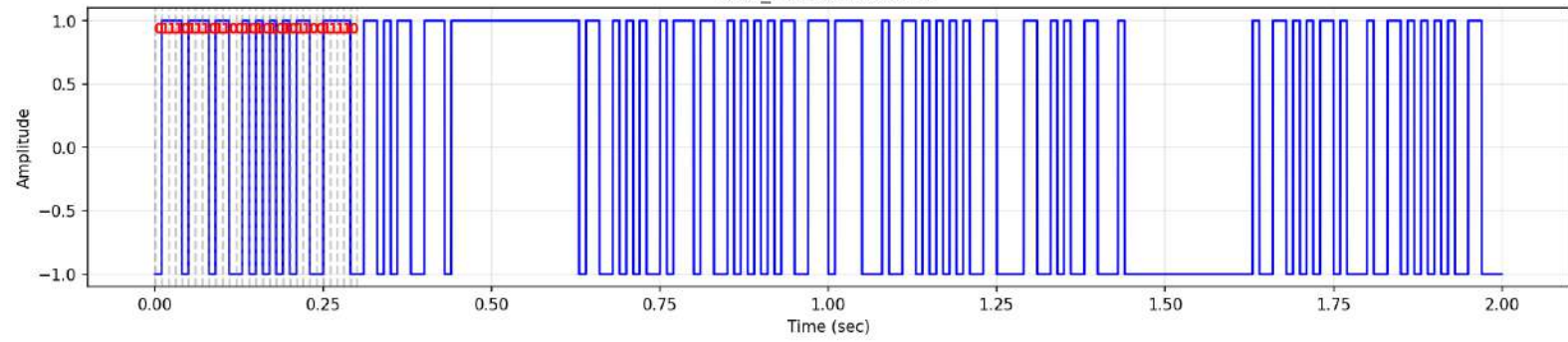


All Line Codes | Bits: [0 1 1 1 0 1 1 1 0 1 1 0 0 1 0 1 0 1 0 1]... | Bit Rate=100 bps

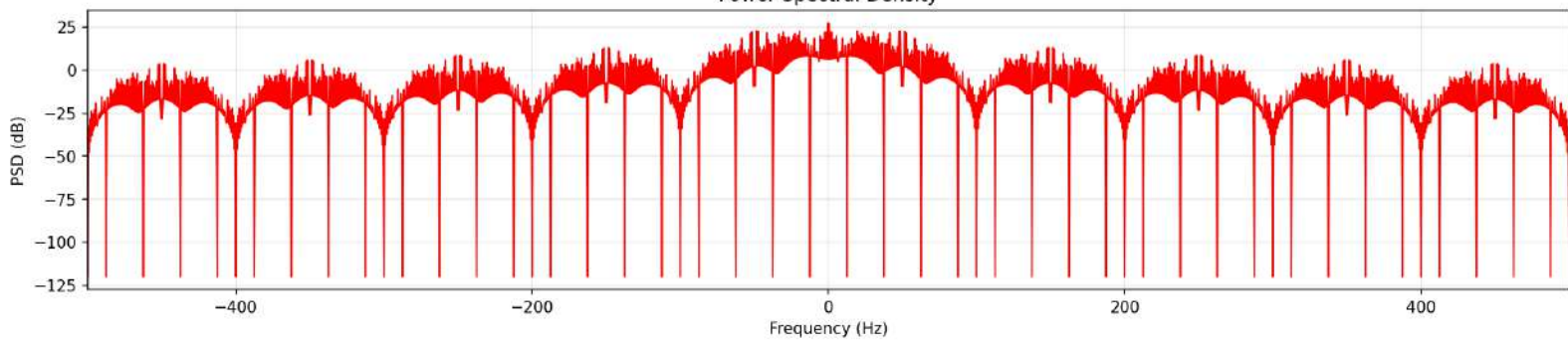


Block 2 | Line Code: NRZ_POLAR | Bit Rate = 100 bps | 30 bits

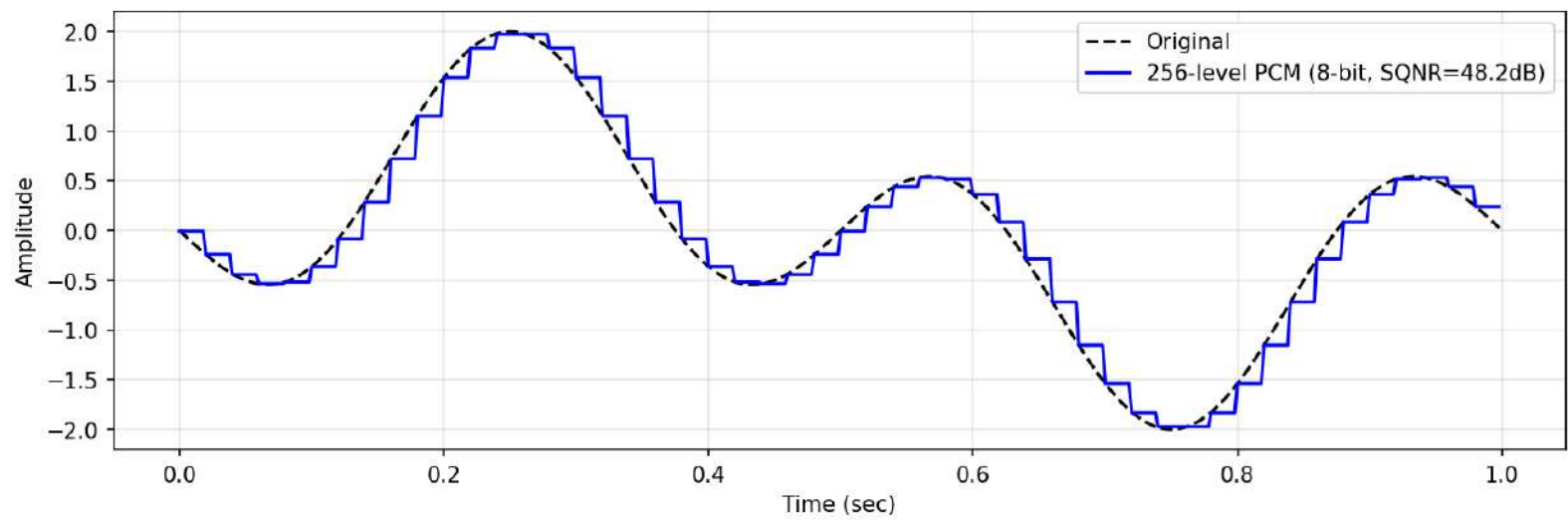
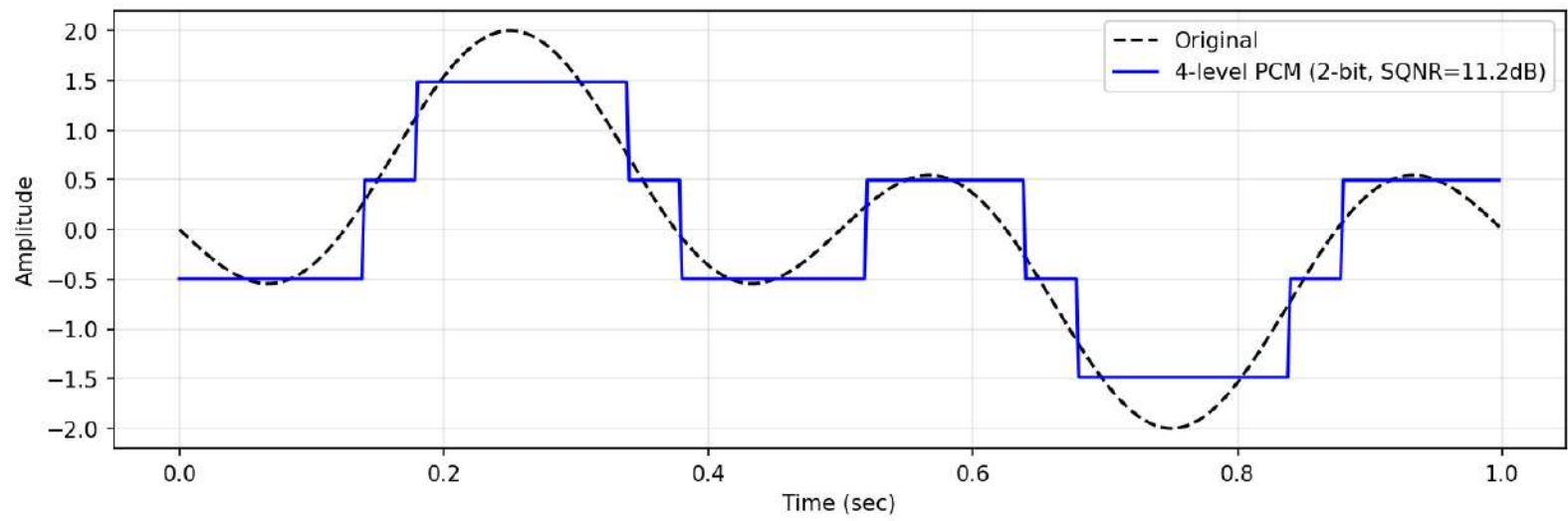
NRZ_POLAR Waveform



Power Spectral Density

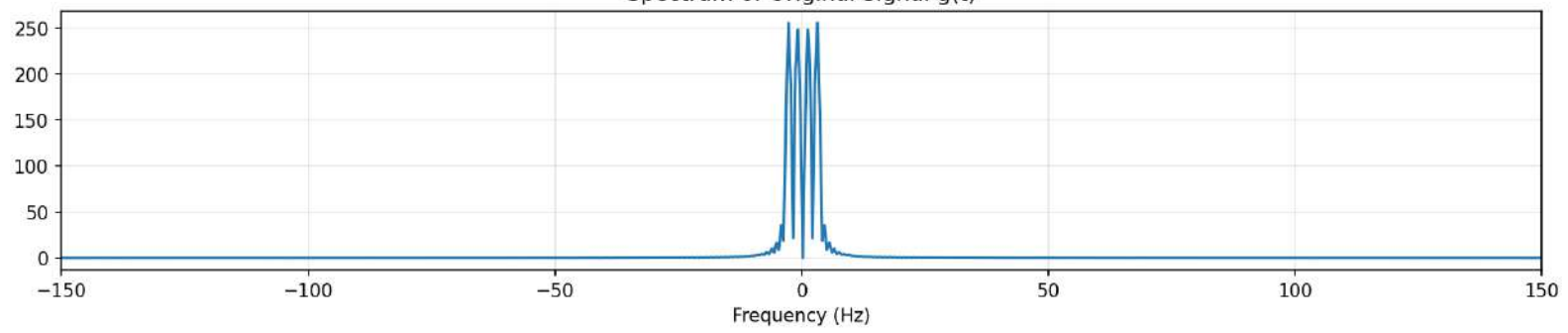


PCM with Different Quantization Levels



Block 1 | Frequency Spectra

Spectrum of Original Signal $g(t)$



Spectrum of Sampled+Quantized Signal ($f_s=50.0$ Hz)

